RAJALAKSHMI ENGINEERING COLLEGE

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A2

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|---|---|
| **Started on** | Wednesday, 26 August 2026, 5:47 PM |
| **State** | Finished |
| **Completed on** | Monday, 31 August 2026, 11:57 AM |
| **Time taken** | 4 days 18 hours |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given two arrays array_One[] and array_Two[] of same size N. We need to first rearrange the arrays such that the sum of the product of pairs( 1 element from each) is minimum. That is SUM (A[i] * B[i]) for all i is minimum.

**For example:**

| Input | Result |
|---|---|
| 3<br>1<br>2<br>3<br>4<br>5<br>6 | 28 |

**Answer:** (penalty regime: 0 %)

```
#include <stdio.h>
#include <stdlib.h>

int ascending(const void *a, const void *b)
{
    int x = *(int *)a;
    int y = *(int *)b;

    if (x < y)
        return -1;
    if (x > y)
        return 1;
    return 0;
}

int descending(const void *a, const void *b)
{
    int x = *(int *)a;
    int y = *(int *)b;

    if (x > y)
        return -1;
    if (x < y)
        return 1;
    return 0;
}

int main()
{
    int n;
    scanf("%d", &n);

    int A[n], B[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &A[i]);

    for (int i = 0; i < n; i++)
        scanf("%d", &B[i]);

    // A in ascending order
    qsort(A, n, sizeof(int), ascending);

    // B in descending order
    qsort(B, n, sizeof(int), descending);

    long long sum = 0;

    for (int i = 0; i < n; i++)
    {
        sum += (long long)A[i] * B[i];
    }
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>1<br>2<br>3<br>4<br>5<br>6 | 28 | 28 | ✔ |
| ✔ | 4<br>7<br>5<br>1<br>2<br>1<br>3<br>4<br>1 | 22 | 22 | ✔ |
| ✔ | 5<br>20<br>10<br>30<br>10<br>40<br>8<br>9<br>4<br>3<br>10 | 590 | 590 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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